



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2016
ASSESSMENT TASK 1

Mathematics Extension 2

General Instructions

- **Working time – 60 minutes**
Reading time – 5 minutes
- Write on both sides of the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators only
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**. (The multiple choice section is answered in a grid within the question booklet)
- Attempt all questions.

Class Teacher:

(Please tick or highlight)

- ☐ Dr Jomaa
- ☐ Mr Ireland
- ☐ Mr Lin

Student Number:

(To be used by the exam markers only.)

Question	1-5	6	7	8	Total	Percent
Mark	$\frac{5}{5}$	$\frac{13}{13}$	$\frac{13}{13}$	$\frac{14}{14}$	$\frac{45}{45}$	$\frac{100}{100}$

Section I

5 Marks

Attempt Questions 1 to 5

Mark your answers on the answer sheet provided.

1. If $z = \frac{3+4i}{1+2i}$, the imaginary part of z is:
A) -2
B) $-\frac{2}{5}i$
C) $-\frac{2}{5}$
D) $-2i$
2. If $z = a + ib$ where a and b are real and non-zero. Which of the following is true?
A) $z + \bar{z}$ is real
B) $\frac{z}{\bar{z}}$ is non real
C) $z^2 - (\bar{z})^2$ is real
D) $z\bar{z}$ is real and positive
3. If $z = \sqrt{3} + i$ then the modulus/argument form $z = 2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$. If $z^n + (\bar{z})^n$ is to be rational, then the integer ' n ' cannot be:
A) 2
B) 3
C) 5
D) 6
4. The polynomial equation $P(x) = 0$ has real coefficients, and has roots which include $x = -2 + i$ and $x = 2$. What is the minimum degree of $P(x)$
(A) 1
(B) 2
(C) 3
(D) 4

5. The polynomial $P(x) = x^3 + 3x^2 - 24x + 28$ has a double zero. What is the value of the double zero?
- (A) -7
(B) -4
(C) 4
(D) 2

END OF SECTION I

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☐ D |

Section II

Total of 40 marks

Attempt Questions 6 to 8.

Write your answers in the writing book provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (13 marks)

- (a) Let $z = 2 + 2i$ and $w = 1 - i$. Find in the form of $x+iy$, where x and y are real,
- (i) $z\bar{w}$ 1
 - (ii) $|z|$ 1
 - (iii) $\arg(zw)$ 2
- (b) Sketch the intersection of the following loci: 3
- $$|z - 2| = 2 \text{ and } -\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{4}$$
- (c) Find the square root of $21 - 20i$ 3
- (d) (i) Express $\frac{-1+i}{\sqrt{2}}$ in modulus-argument form 1
- (ii) Hence express $\left(\frac{-1+i}{\sqrt{2}}\right)^{1002}$ in Cartesian form 2

Question 7 (13 marks)

- (a) It is given that $1 + i$ is a root of $P(z) = 2z^3 - 3z^2 + rz + s$ where r and s are real.
- (i) Explain why $1 - i$ is also a root of the equation. 1
 - (ii) Factorise $P(z)$ over the real field. 2
- (b) The equation $x^3 + px - 1 = 0$ has 3 non zero real roots α, β and γ
- (i) Find the value of $\alpha^2 + \beta^2 + \gamma^2$ 2

(ii) Hence find the value of $\alpha^4 + \beta^4 + \gamma^4$ 2

(iii) Show that p must be negative 1

(iv) Find the monic equation with coefficients in terms of p whose roots are 2

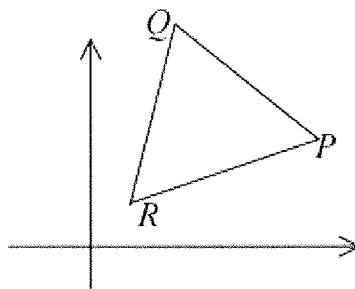
$$\frac{\alpha}{\beta\gamma}, \frac{\beta}{\alpha\gamma}, \frac{\gamma}{\alpha\beta}$$

(c) $P(x) = x^4 + 3x^3 - 6x^2 - 28x + c$ has a zero of multiplicity 3. 3

Find the value of c

Question 8 (14 marks)

(a)



In the Argand diagram above, the points P, Q and R represent the complex numbers p, q and r .

(i) Given that the triangle PQR is equilateral, explain why 1

$$r - q = \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)(q - p)$$

(ii) Hence, or otherwise show that $2r = (p + q) + i\sqrt{3}(q - p)$ 1

(b) (i) Find the roots of $z^5 + 1$, leaving your solutions in modulus-argument form 2

(ii) Prove that the solutions of $z^4 - z^3 + z^2 - z + 1 = 0$ are the non-real solutions of $z^5 + 1 = 0$ 1

(iii) Show that if $z^4 - z^3 + z^2 - z + 1 = 0$ where $z = \cos \theta + i \sin \theta$ then $4 \cos^2 \theta - 2 \cos \theta - 1 = 0$ 3

(iv) Hence, find the exact value of $\sec \frac{3\pi}{5}$ 2

(c) (i) By using $z\bar{z} = |z|^2$ to show that 2

$$\text{Show that } |1 - \bar{z}w|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2)$$

(ii) Deduce that $|1 - \bar{z}w|^2 = |z - w|^2$ if either z or w lies on the unit circle 2

Multiple Choice

1. C
2. C
3. C
4. C
5. D.

Question 6

a) i) $4i$

ii) $2\sqrt{2}$

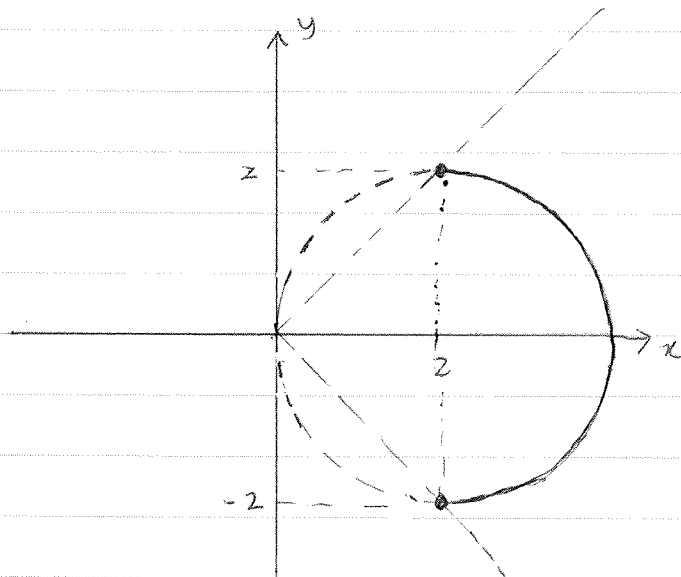
iii) $\arg(zw) = \arg(z) + \arg(w)$

$$= \frac{\pi}{4} - \frac{\pi}{4}$$

$$= 0$$

for correct method.

b)



(intersection describes a semi-circle centre (2,0) and radius of 2.)

c) let $z^2 = 21 - 20i$

if $z = a + ib$

then $a^2 - b^2 = 2$

$$ab = -10$$

$$a^2 + b^2 = 29$$

$\therefore 2a^2 = 50$

$a = \pm 5$ and $b = \mp 2 \quad \therefore$ roots are $\pm (5 - 2i)$

d) i) $\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}$.

ii) $(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})^{1002}$

$$= \cos \left(\frac{1503\pi}{2} \right) + i \sin \left(\frac{1503\pi}{2} \right)$$

$$= 0 - i$$

$$= -i$$

Question 7

a) i) Since $P(z)$ have all real coefficients then roots of $P(z)$ occurs in conjugate pairs.
 $\therefore$ if $1+i$ is a zero then $1-i$ is a zero

ii) From (i) $1+i$ and $1-i$ are zeros. Let α be the third zero of $P(z)$

$$\therefore \text{Sum of zeros are } (1+i) + (1-i) + \alpha = \frac{3}{2}$$

$$\therefore \alpha = -\frac{1}{2}$$

$$\therefore P(z) = 2(z - (1+i))(z - (1-i))(z + \frac{1}{2})$$

$$= (2z+1)(z^2 - 2z + 2)$$

b) i) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

since $\alpha + \beta + \gamma = 0$ (sum of the roots)

and $\alpha\beta + \alpha\gamma + \beta\gamma = p$

$$\alpha^2 + \beta^2 + \gamma^2 = 0 - 2p$$

$$= -2p$$

7 ii) $x^3 = -px + 1$

$\therefore x^4 = -px^2 + x$

since α, β, γ are roots then

$\alpha^4 = -p\alpha^2 + \alpha \quad \text{--- (1)}$

$\beta^4 = -p\beta^2 + \beta \quad \text{--- (2)}$

$\gamma^4 = -p\gamma^2 + \gamma \quad \text{--- (3)}$

Now (1) + (2) + (3) yields

$\alpha^4 + \beta^4 + \gamma^4 = -p(\alpha^2 + \beta^2 + \gamma^2) + (\alpha + \beta + \gamma)$

using part (i)

$$\begin{aligned} \alpha^4 + \beta^4 + \gamma^4 &= -p(-2p) + 0 \\ &= 2p^2 \end{aligned}$$

iii) Since α, β and γ are non-zero and real

then $\alpha^2 + \beta^2 + \gamma^2 > 0$

But since $\alpha^2 + \beta^2 + \gamma^2 = -2p$ (from part (i))

Then $-2p > 0$

$\therefore p < 0$

iv) $\therefore \alpha\beta\gamma = 1$ (product of roots)

the roots $\frac{\alpha}{\beta\gamma}, \frac{\beta}{\alpha\gamma}, \frac{\gamma}{\alpha\beta}$

is equivalent to $\alpha^2, \beta^2, \gamma^2$.

Now replace x with $\sqrt{x}$

$x^{\frac{3}{2}} + p(x)^{\frac{1}{2}} = 1$

$x^{\frac{1}{2}}(x + p) = 1$ and squaring both sides

$x^3 + 2px^2 + p^2x - 1 = 0$

c) $P(x) = x^4 + 3x^3 - 6x^2 - 28x + c$ has zero of multiplicity 3

then if α is the zero

$$\text{then } P(\alpha) = 0$$

$$P'(\alpha) = 0$$

$$P''(\alpha) = 0$$

$$P'(x) = 4x^3 + 9x^2 - 12x - 28$$

$$P''(x) = 12x^2 + 18x - 12$$

$$P''(x) = 0 \Rightarrow 2x^2 + 3x - 2 = 0$$

$$(2x - 1)(x + 2) = 0$$

$$x = \frac{1}{2} \text{ or } x = -2$$

$$P'(-2) = 0$$

$\therefore x = -2$ is a triple zero.

$$P(-2) = 0 \Rightarrow (-2)^4 + 3(-2)^3 - 6(-2)^2 - 28(-2) + c = 0$$

$$14 + c = 0$$

$$c = -14$$

Question 8

a) i) $r - q$ represents the vector $\vec{QR}$

since $\vec{QR}$ is equivalent to $\vec{PQ}$ rotated anticlockwise by $\frac{\pi}{3}$ twice.

(as $\triangle PQR$ is equilateral, and rotating $\vec{PQ}$ twice will form a rhombus, where opposite sides are equal and in same direction)

$$\text{then } r - q = (q - p) \text{cis } \frac{2\pi}{3}$$

ii) From (i) $r - q = (q - p) \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$

$$2r - 2q = -q + \sqrt{3}qi + p - \sqrt{3}pi$$

$$2r = (q + p) + \sqrt{3}(q - p)i$$

b) i) $z^5 = -1$

let $z = r(\cos \theta + i \sin \theta)$

then by De Moivre's Theorem

$$z^5 = r^5 (\cos 5\theta + i \sin 5\theta)$$

and $-1 = \cos(\pi + 2k\pi), \quad k = 0, 1, 2, 3, 4$

$$\therefore 5\theta = \pi + 2k\pi$$

$$\theta = \frac{\pi + 2k\pi}{5}$$

$$\therefore z_1 = \text{cis } \frac{\pi}{5}$$

$$z_2 = \text{cis } \frac{3\pi}{5}$$

$$z_3 = -1$$

$$z_4 = \text{cis } \left(-\frac{3\pi}{5}\right)$$

$$z_5 = \text{cis } \left(-\frac{\pi}{5}\right)$$

$$c) i) |1 - \bar{z}w|^2 - |z - w|^2$$

$$= (1 - \bar{z}w)(\overline{1 - \bar{z}w}) - (z - w)(\overline{z - w}) \quad \text{as } |z|^2 = z\bar{z}$$

$$= (1 - \bar{z}w)(1 - z\bar{w}) - (z - w)(\bar{z} - \bar{w})$$

$$= (1 - z\bar{w} - \bar{z}w + z\bar{z}w\bar{w}) - (z\bar{z} - z\bar{w} - w\bar{z} + w\bar{w})$$

$$= 1 - z\bar{w} - \bar{z}w + |z|^2|w|^2 - |z|^2 + z\bar{w} + w\bar{z} - |w|^2$$

$$= 1 - |z|^2|w|^2 - |z|^2 - |w|^2$$

$$= -|z|^2(1 - |w|^2) + (1 - |w|^2)$$

$$= (1 - |w|^2)(1 - |z|^2)$$

$$ii) \text{ If } |1 - \bar{z}w|^2 = |z - w|^2$$

$$\therefore |1 - \bar{z}w|^2 - |z - w|^2 = 0 \quad \text{then from part (i)}$$

$$(1 - |w|^2)(1 - |z|^2) = 0$$

$$\therefore |w|^2 = 1 \quad \text{or} \quad |z|^2 = 1$$

$$|w| \geq 1, |w| > 0 \quad \text{and} \quad |z| = 1, |w| > 0$$

which lies on the unit circle.

$$ii) \quad z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1) = 0$$

Since $z = -1$ is the real solution then the roots of

$$z^4 - z^3 + z^2 - z + 1 = 0 \quad \text{are the non real roots of}$$

$$z^5 + 1 = 0$$

iii) Dividing through by z^2 , we obtain

$$(z^2 + \frac{1}{z^2}) - (z + \frac{1}{z}) + 1 = 0$$

$$\text{Since } z = \cos \theta + i \sin \theta, \quad \frac{1}{z} = \cos(-\theta) + i \sin(-\theta) \\ = \cos \theta - i \sin \theta$$

$$\therefore z + \frac{1}{z} = 2 \cos \theta$$

$$\text{and similarly } z^2 + \frac{1}{z^2} = 2 \cos 2\theta.$$

$$2(2 \cos^2 \theta - 1) - (2 \cos \theta) + 1 = 0$$

$$4 \cos^2 \theta - 2 - 2 \cos \theta + 1 = 0$$

$$4 \cos^2 \theta - 2 \cos \theta - 1 = 0.$$

$$iv) \quad \text{let } x = \cos \theta$$

$$\therefore 4x^2 - 2x - 1 = 0$$

$$\therefore x = \frac{1 \pm \sqrt{5}}{4} \quad \text{or} \quad \frac{1 - \sqrt{5}}{4}$$

$$\therefore \cos \theta = \frac{1 \pm \sqrt{5}}{4} \quad \text{or} \quad \frac{1 - \sqrt{5}}{4} \quad [1]$$

$$\text{For } \theta = \frac{3\pi}{5}, \quad \cos \theta < 0 \quad [1]$$

$$\therefore \cos \theta = \frac{1 - \sqrt{5}}{4}$$

$$\therefore \sec \theta = \frac{4}{1 - \sqrt{5}}$$